

Activation Functions

David Yuan

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1 SIGMOID

Function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Derivative

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

Pros

- Smooth, differentiable
- Outputs probability like values [0,1]

Cons

- Saturates -> vanishing gradients; for large positive or negative x , $\sigma(x)$ will be equal to 1 or 0, resulting in derivative to be zero, and thus vanishing.
- Not zero-centered
- Computationally expensive vs ReLU

2 TANH

Function

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Derivative

$$\tanh'(x) = 1 - \tanh^2(x)$$

Pros

- Smooth, differentiable
- Zero-centered

Cons

- Saturates -> vanishing gradients; for large positive or negative x , $\tanh(x)$ will be equal to -1 or 1, resulting in derivative to be zero, and thus vanishing.
- Rare use in model deep learning except RNNs

3 RECTIFIED LINEAR UNIT (ReLU)

Function

$$\text{ReLU}(x) = \max(0, x)$$

Derivative

$$\text{ReLU}'(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$$

Pros

- Simple + cheap
- Fast
- Partially solve vanishing gradient

Cons

- "Dead ReLU" - neurons never activate if stuck < 0
- Not smooth at 0

4 LEAKY RELU

Function

$$f(x) = \begin{cases} x & x > 0 \\ \alpha x & x \leq 0 \end{cases}$$

Derivative

$$f'(x) = \begin{cases} 1 & x > 0 \\ \alpha & x \leq 0 \end{cases}$$

Pros

- Fixes "dead ReLU"
- Still cheap

Cons

- α is hyperparameter (typically $\alpha = 0.01$)
- Still non-smooth at 0

5 GELU

Function

$$GELU(x) = x \cdot \Phi(x)$$

where $\Phi(x)$ is normal CDF

Approximation used in practice:

$$GELU(x) = 0.5x \left(1 + \tanh \left(\sqrt{\frac{2}{\pi}} (x + 0.044715x^3) \right) \right)$$

Derivative

- behaves like x for large positive
- behaves like 0 for negative
- smooth transition

Pros

- Better than ReLU for language & attention models
- Keeps small negative values instead of killing them

Cons

- More expensive than ReLU

Uses

- GPT
- BERT
- Most LLMs

6 SWISH/SiLU

Function

$$\text{Swish}(x) = x \cdot \sigma(x)$$

Derivative

$$f'(x) = \sigma(x) + x\sigma(x)(1 - \sigma(x))$$

Pros

- Smooth
- Similar to GELU

Cons

- Slightly slower
- Less used than GELU in LLMs

Uses

- Vision Transformers

7 SOFTMAX

Function

$$s_i(x) = \text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Derivative

$$\frac{\partial s_i}{\partial z_j} = \begin{cases} s_i(1 - s_i) & i = j \\ -s_i s_j & i \neq j \end{cases}$$

Uses

- Language models output layer
- Multi-class outputs